



— ESPRIT · FINAL YEAR PROJECT · 2026



AI-Powered Recruitment Platform

From job discovery to evidence-based hiring decisions.

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The screenshot displays the PEAXIS recruitment platform interface. The main view shows a pipeline for the 'Backend Engineer' role, with 6 candidates in total. The pipeline is divided into two stages: 'Applied' (2 candidates) and 'Screening' (2 candidates). The 'Applied' stage lists Karim Haddad (Full-stack developer, Alignment 62%) and Omar Chaabane (Backend developer, Alignment 71%). The 'Screening' stage lists Amine Benali (Backend engineer, Alignment 85%) and Sarah Mansour (DevOps engineer, Alignment 78%, Reviewed). A callout box highlights Amine Benali's 85% alignment, stating 'Why 85%? traced to CV evidence'. The detailed view of Amine Benali shows his profile (Backend engineer, 4 years, Tunis) and a list of requirements linked to CV evidence: NestJS (Satisfied), PostgreSQL (Satisfied), Docker (Satisfied), 3+ years backend (Satisfied), and Kubernetes (Needs verification). A 'Schedule interview' button is visible at the bottom.



— A SHORT STORY · 1 OF 4



Amine

Backend engineer, looking for a new opportunity

✓ CV submitted

A JOB OFFER

Backend Engineer

Northwind · Tunis · Hybrid

NestJS

PostgreSQL

Docker

✓ It seems to match his experience

cv_amine_benali.pdf



...and waits...



— A SHORT STORY · 2 OF 4



The recruiter

Applications keep coming in

AB

Amine Benali
Backend engineer · Tunis

85%

match

Should Amine move forward?

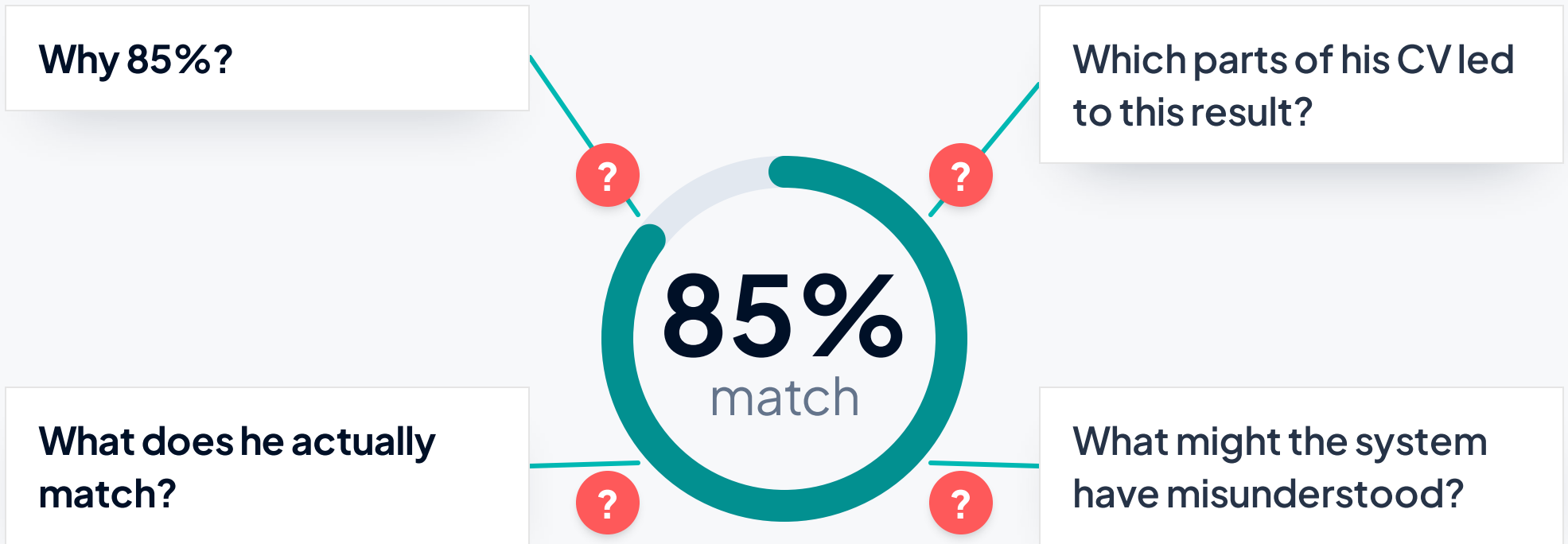
Invite to interview

Reject



— A SHORT STORY · 3 OF 4

Before deciding, several questions remain





— A SHORT STORY · 4 OF 4

The real question is not whether the result is ~~85%~~.

Can we trust a number if we don't understand how we got there?



— INTRODUCTION

Presentation plan

From the recruitment problem to a working product.

01	Context & problem	Why recruitment needs clearer decisions
02	Methodology & requirements	How I organized and defined the project
03	The PEAXIS platform	The candidate and recruiter journey
04	Architecture, AI & production	How the system works and stays reliable
05	Results, perspectives & live demo	What exists today and what comes next



CONTEXT

Built at **Prospecter**, an AI product company

THE HOST COMPANY

AI product for B2B prospecting, based in Doha, Qatar.

I worked there full time as a software engineer: AI, software, and a real product in production.

MY WORK COVERED THREE AREAS

 **The application**
Web platform, API, business rules

 **The AI**
LLM integration, evidence assessment

 **The infrastructure**
Deployment, security, operations

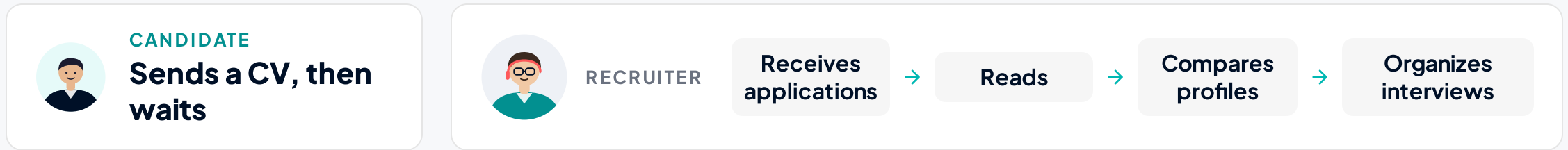
Built independently, from architecture to deployment.

Supervised by **Mme Olfa Mannai** (academic) and **Mr. Fedi Naimi** (company). Thank you for your support.



PROBLEM

Hiring, seen from both sides



📄 When applications grow, the work becomes hard to manage

 **≈ 1.5 months** median time to fill a role (SHRM)

 **37%** of recruiting teams already try generative AI (LinkedIn)

AI can help with the workload, but a hiring decision still needs a reason people can understand.



PROBLEM

Three problems to solve

01



Manual work

Reading and comparing CVs takes a lot of time.

02



Scattered tools

The CV is in one place, the discussion in email, the interview in a calendar.

03



Unclear AI results

A score appears, but the recruiter cannot see why.

These are the three problems I wanted PEAXIS to solve.



PROBLEM

Existing solutions, and our difference

SOLUTION

STRONG AT

PEAXIS FOCUS

LinkedIn Recruiter

+ Hiring Assistant

Sourcing: finding candidates in a huge network, AI shortlists

Starts after the application, with evidence per requirement

Greenhouse

Structured hiring

Mature enterprise ATS, AI answers linked to sources, human-led

Same spirit, lighter product, evidence in every candidate review

Ashby

All-in-one suite

ATS + CRM + scheduling + analytics, AI throughout

Narrower on purpose: centred on explainable assessment

Workable

SME-friendly ATS

Quick to start, AI agent, paid plans from \$299/month

Free entry plan, billing in local currency (TND)

PEAXIS: one candidate-to-recruiter journey, every AI result linked to evidence, the human decides.



PROBLEM · PROJECT OBJECTIVE

How can AI make recruitment more efficient, while keeping decisions understandable?



Bring the workflow
together



Understand the
candidate's
information



Assist the recruiter



Keep the human in
control



METHODOLOGY

Scrum, adapted to working alone

01

Backlog

What to build, in order



02

Sprint

Build something that works



03

Review

Check the result



04

Retrospective

Improve how I work



Tracked in the repository

Epics, tasks and review checklists



Verified automatically

Automated checks on every change

Then I started with the users: what does each one need to do?



REQUIREMENTS

Functional requirements

Candidate

- ✓ Discover jobs
- ✓ Build a profile from a CV
- ✓ Apply in a few clicks
- ✓ Follow every application

Recruiter

- ✓ Create and publish jobs
- ✓ Manage candidates
- ✓ Run the hiring pipeline
- ✓ Review evidence
- ✓ Schedule interviews
- ✓ Use AI assistance

Platform

- ✓ Multi-tenant organizations
- ✓ Roles and access
- ✓ Billing and entitlements
- ✓ Protected files
- ✓ Notifications

PLATFORM

Each company is an isolated organization, with roles, plans and entitlements, private files and notifications.



REQUIREMENTS

Non-functional requirements



Security

Data isolated per organization



Reliability

No loss of business data



Performance

Responsive screens



Scalability

Grow with usage



Explainability

Results can be justified



Maintainability

Easy to evolve safely

MAINTAINABILITY

Typed code, automated tests and CI, and documented architecture decisions.



— SOLUTION

PEAXIS: two experiences, one platform



PEAXIS Jobs

For candidates

Discover jobs · profile from CV · apply · track



PEAXIS Hire

For hiring teams

Jobs · pipeline · evidence · interviews · AI assistance

SHARED PLATFORM

Identity

Organizations

Permissions

Files

Billing

AI



SOLUTION

PEAXIS Jobs: the candidate experience

AB Following Amine

01 Discover

02 Job details

03 Apply with CV

04 Track application

peaxis.com/jobs

My applications

N

Backend Engineer
Northwind · Applied today

Under review

✓

Applied

2

Screening

3

Interview

4

Offer

L

DevOps Engineer
Lumen Labs · Applied 5 days ago

Interview



SOLUTION

PEAXIS Hire: the recruiter workspace

AB Following Amine

northwind.peaxis.com/pipeline

Backend Engineer

Pipeline · 6 candidates

KanbanList

Applied2

KH Karim Haddad
Full-stack developer

Alignment 62%

OC Omar Chaabane
Backend developer

Alignment 71%

Screening1

SM Sarah Mansour
DevOps engineer

Alignment 78%

✓ Reviewed

Interview2

LG Lina Gharbi
Backend engineer

Alignment 90%

✓ Reviewed

AB Amine Benali
Backend engineer · 4 yrs

Alignment 85%

Interview · Thu 10:00

Offer0

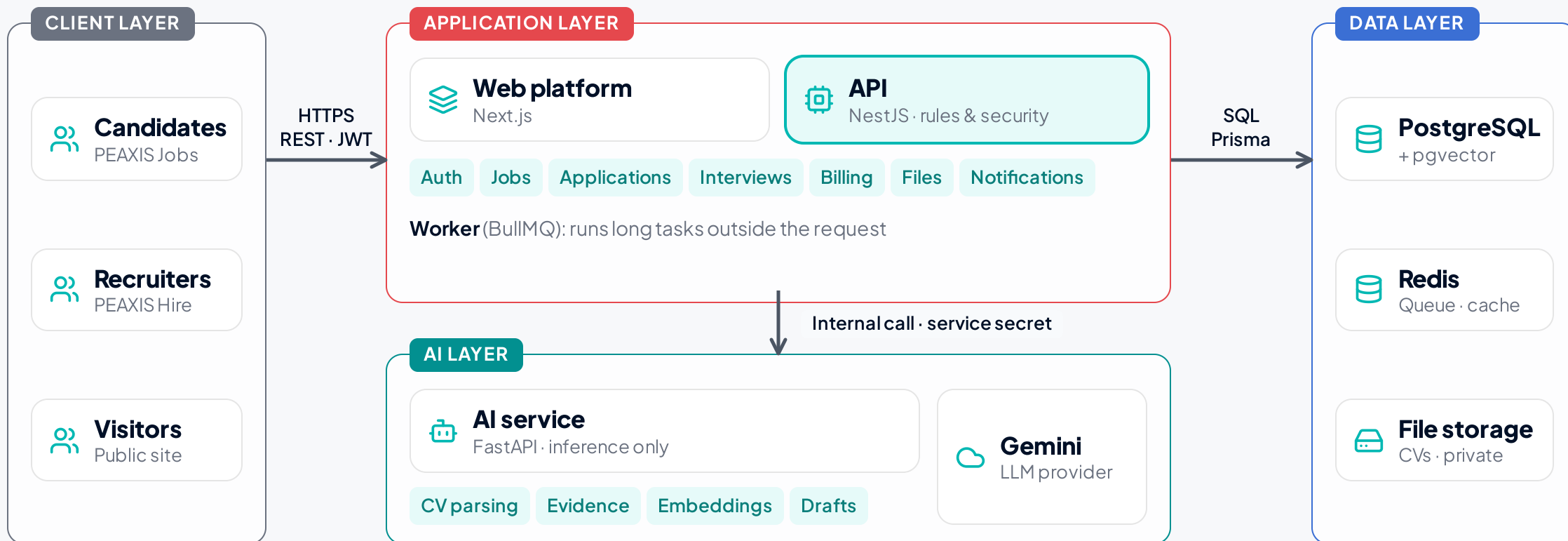
17 / 31 · SOLUTION

Reconstructed from the PEAXIS interface · fictitious data



— ARCHITECTURE

Logical architecture

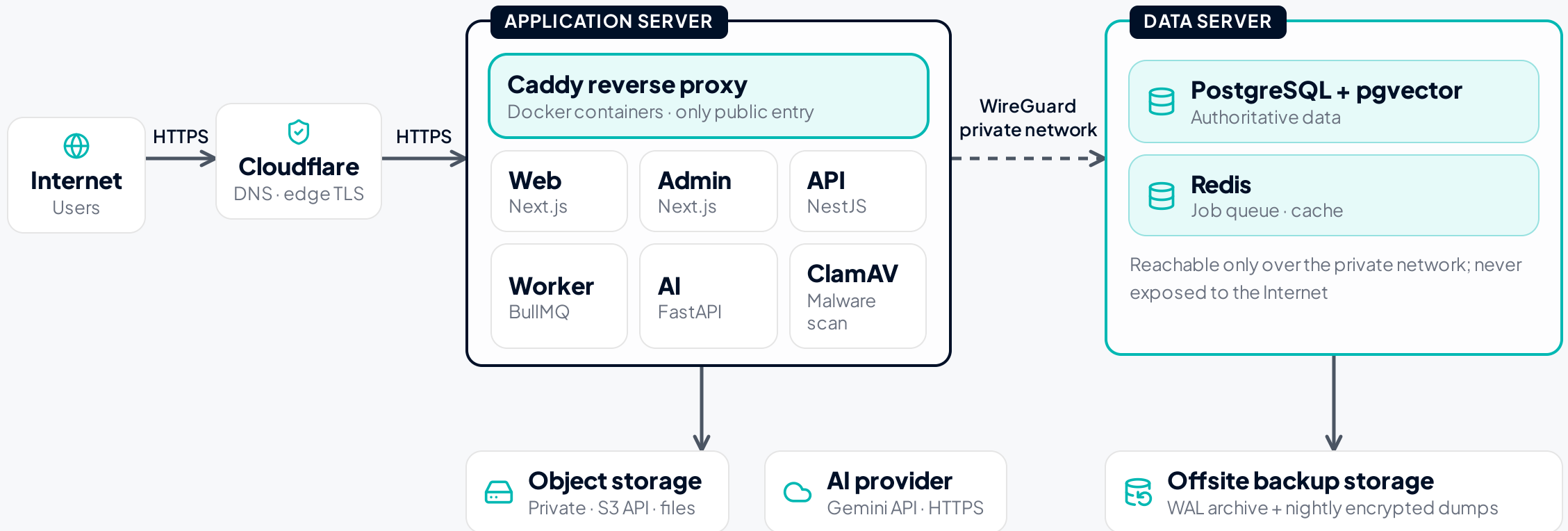


The API is the single authority: the AI layer never touches the data layer.



ARCHITECTURE

Physical architecture

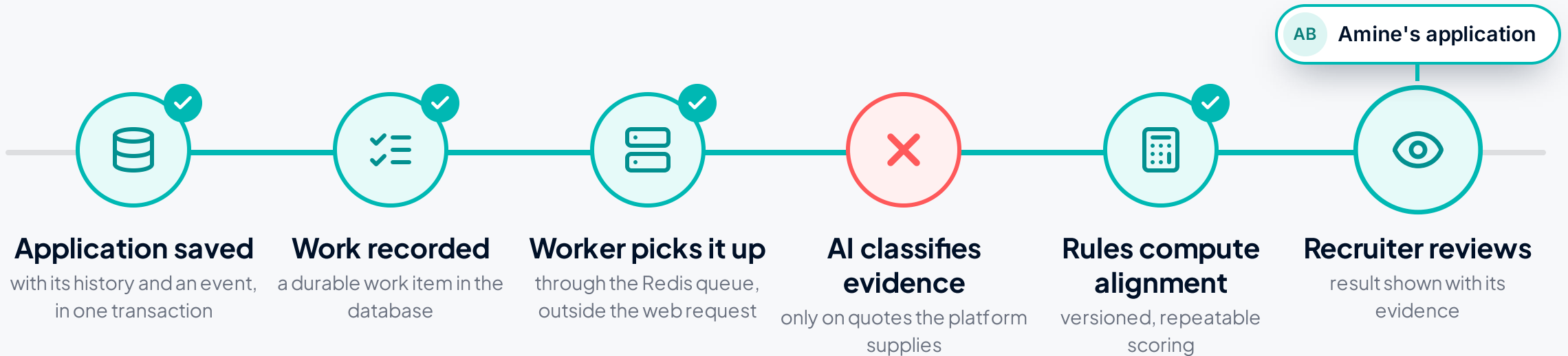


The public application and the data layer are separated; the database and cache are never exposed to the Internet.



— ARCHITECTURE

What happens when Amine applies



What if the AI service is down?



Amine's application is still saved. The work is retried later.



AI

How PEAXIS understands a CV

AMINE BENALI

Backend Engineer · Tunis

EXPERIENCE

Backend Engineer · Northwind, 2021 – 2025

- Built REST APIs with **NestJS** serving 40k users
- Designed **PostgreSQL** schemas and migrations
- Containerized services with **Docker**

SKILLS

TypeScript, NestJS, PostgreSQL, Docker, CI/CD

EDUCATION

Software Engineering degree, 2021



Candidate profile

✓ Checked against source

ROLE	Backend Engineer	✓
EXPERIENCE	Northwind · 2021 – 2025 (4 years)	✓
SKILLS	NestJS PostgreSQL Docker TypeScript	✓
EDUCATION	Software Engineering · 2021	✓

Extract content



AI structures information



Validate against source



Profile & evidence

PEAXIS does not blindly trust generated output. Parsed information is checked against the source before it becomes usable evidence.



— AI

Why 85%?

JOB REQUIREMENTS

NestJS Required

PostgreSQL Required

Docker Required

3+ years backend Required

Kubernetes Nice to have

CANDIDATE EVIDENCE

→ *“Built REST APIs with NestJS...”*

→ *“Designed PostgreSQL schemas...”*

→ *“Containerized services with Docker...”*

→ *“Backend engineer, 2021 – 2025”*

→ *No evidence found in the CV*

ASSESSMENT

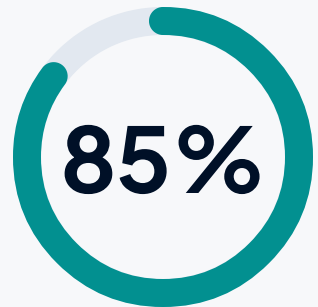
→ ✓ Satisfied

→ ✓ Satisfied

→ ✓ Satisfied

→ ✓ Satisfied

→ ⓘ Needs verification



ALIGNMENT

Every result can be traced back to evidence.

AI helps retrieve and classify relevant evidence. The platform applies the final assessment rules.

Must-have skills weigh most, then experience, then nice-to-haves. A missing must-have caps the score, so a high number is never misleading.



— AI

Explainable, **human-controlled** AI

AI ASSISTS

Parse

Summarize

Explain

Recommend

Draft

HUMANS CONTROL

Requirements

Interviews

Stage changes

Rejections

Offers

Show the evidence. Let the human decide.

AI supports recruiter judgment, it does not replace it. Evidence-linked, reviewable, advisory only: no AI step moves a candidate or makes an offer.



PRODUCTION

Engineered to be **secure, recoverable, operable**

Security

- Tenant isolation: data stays within its organization
- Secure access: hashed passwords, token sessions, MFA for platform admins
- Private data: protected files, scanned uploads
- Protected infrastructure: private database network, firewall

Reliability

- Durable background work with retries
- Events saved with the business change
- An AI outage never loses an application
- Backups with a restore procedure

Scalability

- Web, API, worker and AI are separate
- Long AI work stays off the request path
- Workers can grow independently
- Files live outside the app server

Operations

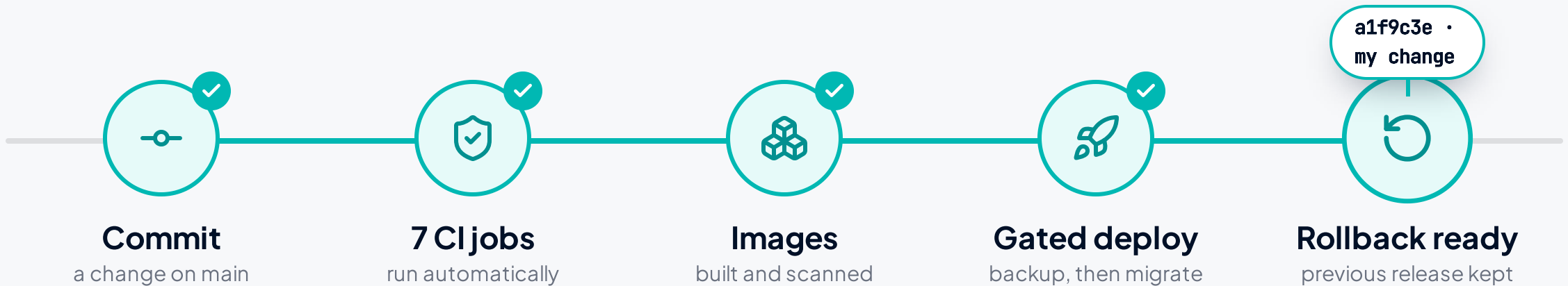
- Health checks and worker heartbeat
- Structured logs, metrics, dashboards
- Alerting through Grafana Cloud
- Next hardening: routine restore drills

We did not only build features: we designed how PEAXIS stays secure, recoverable and operable. No Kubernetes needed at today's scale; clear boundaries for tomorrow's.



— PRODUCTION

From commit to production, safely



WHAT THE CI JOBS CHECK

Lint and types

1,400+ tests

Migrations + end-to-end

Secret scan

Image scan

Dependency audit

Nothing reaches production without passing the checks, and every release can be rolled back.



PRODUCTION

We can see what the system is doing

COLLECTED FROM

Application server
Data server

node exporter · cAdvisor · Caddy ·
PostgreSQL & Redis exporters



Grafana Alloy

One agent per server



Grafana Cloud

 Prometheus

 Loki logs

 Dashboards

 Alerts

Plus synthetic checks on the public site and API.



Servers & containers

CPU, memory, disk, network



API & AI service

Errors, latency, provider calls



Queue & worker

Backlog, failures, heartbeat



Data & backups

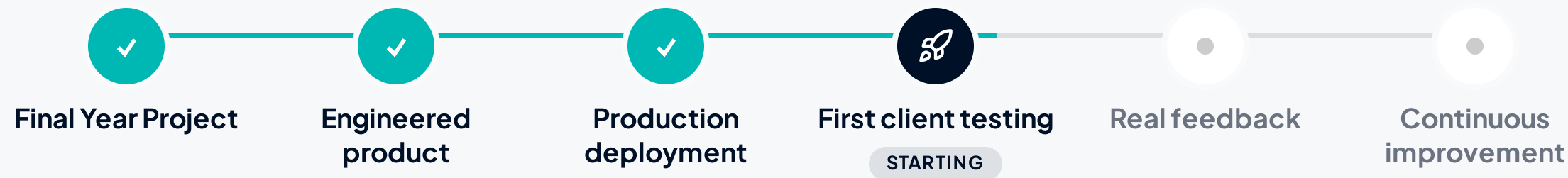
PostgreSQL, Redis, backup status

My next step: verify the alert delivery at every release.



— PRODUCTION

From Final Year Project to **real product**



1,400+

automated tests

API, web, admin and AI service

27

backend modules

122 data models, 28 migrations

7

automated CI jobs

quality, tests, security scans

68/68

production checks passed

at the first production deployment

PEAXIS was not built only to pass a school project. It is a SaaS product for real companies, and the first ones are starting to test it.



— PERSPECTIVES

Where PEAXIS goes next

DIRECTION 1

Deepen PEAXIS Hire

Collaboration

Scorecards, feedback, shared ownership

Communication hub

Candidate email and invitations in one place

Source-linked copilot

Comparisons, interview kits, evidence gaps

Enterprise readiness

Fine-grained permissions, integrations, analytics



DIRECTION 2 **FUTURE**

PEAXIS HR

Beyond hiring: the employee lifecycle. Possible areas, not commitments:

Employee records

Leave & attendance

Documents

Performance

Employee lifecycle

These are future directions. Today, my focus is the recruitment platform you just saw.



LIVE DEMO · ABOUT 5 MINUTES

PEAXIS in action

CANDIDATE

Discover a job



Apply with CV

RECRUITER

Application received



Review candidate



Understand evidence



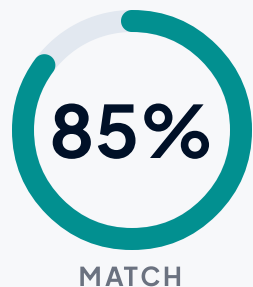
Continue hiring workflow

SWITCH TO THE APP · PRESS → TO RETURN



CONCLUSION

Back to our first question: would you invite a candidate with an 85% match?



Now we have something better than a number: **the evidence behind it.**

NestJS

✓ Satisfied

PostgreSQL

✓ Satisfied

Docker

✓ Satisfied

3+ years backend

✓ Satisfied

Kubernetes

⚠ Needs verification

THAT IS WHAT I BUILT WITH PEAXIS



One recruitment platform

Jobs for candidates, Hire for recruiters



Explainable AI

Every result linked to evidence



Production engineering

Built to run for real users

Show the evidence. **Let the human decide.**



— THANK YOU

Questions?

Thank you to the jury, and to everyone here, for your attention.

BIBANI Mohamed Habib Allah

PEAXIS · Final Year Project · ESPRIT 2026

Show the evidence. Let the human decide.